

Nama : Achmad Pradita Dwi Firmansyah
NIM : 254107020130
Kelas / Absen : TI = 1G / 01

JOBSHEET XIV

Collection

13.1 Praktikum - Implementasi ArrayList

Kode Program

Class Customer

```
package Praktikum14;

public class Customer {
    public int id;
    public String name;

    Customer() {

    }

    Customer(int id, String name) {
        this.id = id;
        this.name = name;
    }

    public String toString(){
        return "ID: " +this.id+ " Nama: " +this.name;
    }
}
```

Class Book

```
package Praktikum14;

public class Book {
    String isbn;
    String title;

    Book() {

    }

    Book(String isbn, String title) {
        this.isbn = isbn;
        this.title = title;
    }

    public String toString() {
        return "ISBN: " +this.isbn+" Title: " +this.title;
    }
}
```

Class DemoArrayList

```
package Praktikum14;
import java.util.ArrayList;
public class DemoArrayList {
    public static void main(String[] args) {
        ArrayList<Customer> customers = new ArrayList<>(2);

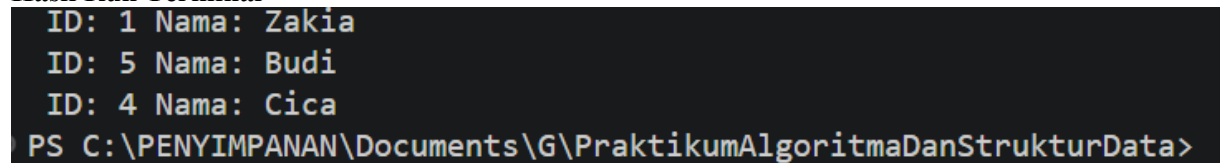
        Customer customer1 = new Customer(1, "Zakia");
        Customer customer2 = new Customer(5, "Budi");

        customers.add(customer1);
        customers.add(customer2);

        customers.add(new Customer(4, "Cica"));

        for (Customer cust : customers) {
            System.out.println(cust.toString());
        }
    }
}
```

Hasil Run Terminal



```
ID: 1 Nama: Zakia
ID: 5 Nama: Budi
ID: 4 Nama: Cica
PS C:\PENYIMPANAN\Documents\G\PraktikumAlgoritmaDanStrukturData>
```

1. Di mana object yang baru ditambahkan? Di awal, di tengah, atau di akhir collection?
Object yang baru ditambahkan di akhir collection

Class DemoArrayList

```
package Praktikum14;
import java.util.ArrayList;
public class DemoArrayList {
    public static void main(String[] args) {
        ArrayList<Customer> customers = new ArrayList<>(2);

        Customer customer1 = new Customer(1, "Zakia");
        Customer customer2 = new Customer(5, "Budi");

        customers.add(customer1);
        customers.add(customer2);

        customers.add(new Customer(4, "Cica"));

        customers.add(2, new Customer(100, "Rosa"));

        for (Customer cust : customers) {
            System.out.println(cust.toString());
        }
    }
}
```

```
}
```

```
ID: 1 Nama: Zakia
ID: 5 Nama: Budi
ID: 100 Nama: Rosa
ID: 4 Nama: Cica
PS C:\PENYIMPANAN\Documents\G\PraktikumAlgoritmaDanStrukturData>
```

2. Index pada ArrayList dimulai dari 0 atau 1?

Index pada ArrayList dimulai dari 0

Class DemoArrayList

```
package Praktikum14;
import java.util.ArrayList;
public class DemoArrayList {
    public static void main(String[] args) {
        ArrayList<Customer> customers = new ArrayList<>();

        Customer customer1 = new Customer(1, "Zakia");
        Customer customer2 = new Customer(5, "Budi");

        customers.add(customer1);
        customers.add(customer2);

        customers.add(new Customer(4, "Cica"));

        customers.add(2, new Customer(100, "Rosa"));

        System.out.println(customers.indexOf(customer2));

        Customer customer = customers.get(1);
        System.out.println(customer.name);
        customer.name = "Budi Utomo";

        ArrayList<Customer> newCustomers = new ArrayList<>();
        newCustomers.add(new Customer(201, "Della"));
        newCustomers.add(new Customer(202, "Victor"));
        newCustomers.add(new Customer(203, "Sarah"));

        customers.addAll(newCustomers);

        System.out.println(customers);
    }
}
```

```
1
Budi
[ ID: 1 Nama: Zakia, ID: 5 Nama: Budi Utomo, ID: 100 Nama: Rosa, ID: 4 Nama: Cica, ID: 201 Nama: Della, ID: 202 Nama:
Victor, ID: 203 Nama: Sarah]
PS C:\PENYIMPANAN\Documents\G\PraktikumAlgoritmaDanStrukturData>
```

3. Cobalah hapus angka 2 saat instansiasi object customers. Apakah ArrayList dapat diinstansiasi tanpa harus menentukan size di awal?

Menghapus angka 2 saat instansiasi object customers, ArrayList tetap dapat diinstansiasi tanpa harus menentukan size di awal

13.2 Praktikum - Implementasi Stack

Class StackDemo

```
package Praktikum14;
import java.util.Stack;
public class StackDemo {
    public static void main(String[] args) {
        Book book1 = new Book("1234", "Dasar Pemograman");
        Book book2 = new Book("7145", "Hafalah Shalat Delisa");
        Book book3 = new Book("9562", "Muhammad Al-Fatih");

        Stack<Book> books = new Stack<>();
        books.push(book1);
        books.push(book2);
        books.push(book3);

        Book temp = books.peek();
        if (temp != null) {
            System.out.println(temp.toString());
        }

        Book temp2 = books.pop();
        if (temp2 != null) {
            System.out.println(temp2.toString());
        }
    }
}
```

1. Mengapa perlu ada pengecekan (temp != null)?

Agar program hanya bisa mencetak data buku ketika nilai temp tidak kosong, karena jika kosong program akan error, sehingga perlu dilakukan pengecekan temp != null agar jika nilai temp kosong program tidak akan menjalankan perintah di dalamnya.

Class StackDemo

```
package Praktikum14;
import java.util.Stack;
public class StackDemo {
    public static void main(String[] args) {
        Book book1 = new Book("1234", "Dasar Pemograman");
        Book book2 = new Book("7145", "Hafalah Shalat Delisa");
        Book book3 = new Book("9562", "Muhammad Al-Fatih");

        Stack<Book> books = new Stack<>();
        books.push(book1);
        books.push(book2);
        books.push(book3);

        Book temp = books.peek();
        if (temp != null) {
            System.out.println(temp.toString());
        }

        Book temp2 = books.pop();
        if (temp2 != null) {
            System.out.println(temp2.toString());
        }
    }
}
```

```

        for (Book book : books) {
            System.out.println(book.toString());
        }

        System.out.println(books);
    }
}
ISBN: 9562 Title: Muhammad Al-Fatih
ISBN: 9562 Title: Muhammad Al-Fatih
ISBN: 1234 Title: Dasar Pemograman
ISBN: 7145 Title: Hafalah Shalat Delisa
[ISBN: 1234 Title: Dasar Pemograman, ISBN: 7145 Title: Hafalah Shalat Delisa]
PS C:\PENYIMPANAN\Documents\G\PraktikumAlgoritmaDanStrukturData>

```

2. Bagaimana cara melakukan pencarian elemen pada stack menggunakan method search()?

```

int bk = books.search(book1);
System.out.println(book1 +" Ada di indeks : "+bk);
ISBN: 1234 Title: Dasar Pemograman Ada di indeks : 2
PS C:\PENYIMPANAN\Documents\G\PraktikumAlgoritmaDanStrukturData>

```

book1 ada di indeks 2 karena penghitungan di mulai dari atas

13.3 Praktikum - Implementasi TreeSet

```

package Praktikum14;
import java.util.TreeSet;
public class TreeSetDemo {
    public static void main(String[] args) {
        TreeSet<String> fruits = new TreeSet<>();

        fruits.add("Mangga");
        fruits.add("Apel");
        fruits.add("Jeruk");
        fruits.add("Jambu");

        for (String temp : fruits) {
            System.out.println(temp);
        }
    }
}
Apel
Jambu
Jeruk
Mangga
PS C:\PENYIMPANAN\Documents\G\PraktikumAlgoritmaDanStrukturData>

```

1. Mengapa urutan yang ditampilkan berbeda dengan urutan penambahan data ke dalam TreeSet fruits?

Karena Tree otomatis mengurutkan dari huruf paling kecil sehingga ketika penambahan data dimulai dari Mangga → Apel → Jeruk → Jambu urutannya berbeda ketika menampilkan tree yaitu Apel → Jambu → Jeruk → Mangga

Class TreeSetDemo

```

package Praktikum14;
import java.util.TreeSet;
public class TreeSetDemo {
    public static void main(String[] args) {
        TreeSet<String> fruits = new TreeSet<>();

        fruits.add("Mangga");
        fruits.add("Apel");
        fruits.add("Jeruk");
        fruits.add("Jambu");

        for (String temp : fruits) {
            System.out.println(temp);
        }

        System.out.println("First: "+fruits.first());
        System.out.println("Last: "+fruits.last());

        fruits.remove("Jeruk");
        System.out.println("Setelah remove " +fruits);

        fruits.pollFirst();
        System.out.println("Setelah pool first " +fruits);

        fruits.pollLast();
        System.out.println("Setelah pool last "+fruits);
    }
}

```

```

Apel
Jambu
Jeruk
Mangga
First: Apel
Last: Mangga
Setelah remove [Apel, Jambu, Mangga]
Setelah pool first [Jambu, Mangga]
Setelah pool last [Jambu]
PS C:\PENYIMPANAN\Documents\G\PraktikumAlgoritmaDanStrukturData>

```

2. Apa yang dilakukan oleh method first(), last(), remove(), pollFirst(), dan pollLast()?
 - Method first() : akan mengambil data paling awal yaitu Apel
 - Method last() : akan mengambil data terakhir yaitu Mangga
 - Method remove() : menghapus data yang sesuai dengan data yang ingin dihapus
 - Method pollFirst() : mengambil data paling awal sekaligus menghapus data paling awal
 - Method pollLast() : mengambil data terakhir sekaligus menghapus data paling akhir

13.4 Praktikum – Sorting

```

package Praktikum14;
import java.util.ArrayList;
import java.util.Collections;
public class DemoArrayList {
    public static void main(String[] args) {
        ArrayList<Customer> customers = new ArrayList<>();
    }
}

```

```

Customer customer1 = new Customer(1, "Zakia");
Customer customer2 = new Customer(5, "Budi");

customers.add(customer1);
customers.add(customer2);

customers.add(new Customer(4, "Cica"));

customers.add(2, new Customer(100, "Rosa"));

System.out.println(customers.indexOf(customer2));

Customer customer = customers.get(1);
System.out.println(customer.name);
customer.name = "Budi Utomo";

ArrayList<Customer> newCustomers = new ArrayList<>();
newCustomers.add(new Customer(201, "Della"));
newCustomers.add(new Customer(202, "Victor"));
newCustomers.add(new Customer(203, "Sarah"));

customers.addAll(newCustomers);

System.out.println(customers);

ArrayList<String> daftarSiswa = new ArrayList<>();
daftarSiswa.add("Zainab");
daftarSiswa.add("Andi");
daftarSiswa.add("Rara");
Collections.sort(daftarSiswa);

System.out.println(daftarSiswa);

customers.sort((c1,c2) -> c1.name.compareTo(c2.name));
System.out.println(customers);
}
}
1
Budi
[ ID: 1 Nama: Zakia, ID: 5 Nama: Budi Utomo, ID: 100 Nama: Rosa, ID: 4 Nama: Cica, ID: 201 Nama: Della, ID: 202 Nama: Victor, ID: 203 Nama: Sarah]
[Andi, Rara, Zainab]
[ ID: 5 Nama: Budi Utomo, ID: 4 Nama: Cica, ID: 201 Nama: Della, ID: 100 Nama: Rosa, ID: 203 Nama: Sarah, ID: 202 Nama: Victor, ID: 1 Nama: Zakia]
PS C:\PENYIMPANAN\Documents\G\PraktikumAlgoritmaDanStrukturData>

```